PRACTICE SHEET

1. Consider the following statements:
(i)The function f (x) = greatest integer ≤ x, x ∈ R is a continuous function.
(ii)All trigonometric functions are continuous on R.
Which of the statements given above is / are correct?
(a) 1 Only (b) 2 Only
(c) Both 1 and 2 (d) Neither 1 nor 2

2. If $f(x)=\begin{cases} mx+1 & x\le\frac{\pi}{2} \\ \sin x+n & x>\frac{\pi}{2}\end{cases}$ is continuous at $x\le\frac{\pi}{2}$, then which one of the following is correct?

(a) m = 1, n = 0 (b) $m=\frac{n\pi}{2}+1$

(c) $n=m\left(\frac{\pi}{2}\right)$ (d) $m=n=\frac{\pi}{2}$

3. If f (x) = $(x+1)^{\cot x}$ is continuous at x = 0, then what is f (0) equal to?
(a) 1 (b) e
(c) 1/e (d) e^2

4. If the derivative of the function

$f(x)=\begin{cases} ax^2+b & x<-1 \\ bx^2+ax+4 & x\ge -1\end{cases}$

9. Consider the following function f:R→such that f(x) = x, if x ≥ 0 and f(x) = – x^2, if x < 0 then, which one of the following is correct?
(a)f(x) is continuous at every x∈R
(b)f(x) is continuous at x = 0 only
(c)f(x) is discontinuous at x = 0 only
(d)f(x) is discontinuous at every x∈R

10. The set of points where the function f(x)=x|x| is differentiable is:
(a) (–∞, ∞) (b) (–∞, 0)∪(0,∞)
(c) (0, ∞) (d) [0, ∞]

11. Consider the following statements:

I. $\lim\limits_{x\to 0}\frac{x^2}{x}$ exists.

II. $\left(\frac{x^2}{x}\right)$ is not continuous at x = 0.

III. $\lim\limits_{x\to 0}\frac{|x|}{x}$ does not exist

Which of the above statements are correct?
(a) I,II and III (b) I and II
(c)II and III (d) I and III

12. **Assertion (A):** $(x)=\begin{cases} x^2\sin\frac{1}{x}, & x\ne 0 \\ 0, & x=0\end{cases}$ is continuous at x = 0